3ME41A(ME): ENGINEERING THERMODYNAMICS

Quiz (M.M. 5)
Each question carries one mark.

1. When a gas is throttled, the maximum temperature drop or cooling will occur if the initial state lies on the
   (a) Saturation curve
   (b) inversion curve
   (c) constant enthalpy curve
   (d) heating region
2. The vapour pressure of a liquid at any arbitrary temperature can be estimated approximately with the help of
   (a) Gibbs equation
   (b) Joule-Kelvin equation
   (c) Clausius-Clapeyron equation
   (d) Maxwell equation
3. Cylinder clearance in a compressor should be
   (a) as large as possible
   (b) as small as possible
   (c) about 50% of swept volume
   (d) about 100% of swept volume
4. The specific ~~gas~~ heat of an ideal gas depends on its
   (a) Density
   (b) Molecular weight
   (c) Pressure
   (d) Temperature
5. Volumetric efficiency is
   (a) the ratio of stroke volume to clearance volume
   (b) the ratio of the air actually delivered to the amount of piston displacement
   (c) reciprocal of compression ratio
   (d) index of compressor performance

MBM University, Jodhpur
Department of Mechanical Engineering
B.E. IInd Year IIIrd Semester
Second Mid-Term Examination Session 2023-24
ME-203A: Kinematics of Machines

Time: 1 hour **Maximum Marks – 20**

Q. 1(a). In a four bar mechanism ABCD, the link AD is fixed link of 120 mm long, crank AB is 30 mm long and rotate uniformly at 100 rpm clockwise, while the link CD is 60 mm long and oscillates about fixed point D. The link BC is equal to length AD. Find the angular velocity of link DC when angle BAD is 60°. (6)

(b) What do you mean by instantaneous centre? Explain the properties of instantaneous centre. (4)

Q. 2 What is the difference between lower and higher pairs? Give examples of each type. (3)

(b) Sketch and explain the various inversion of single-slider crank chain mechanism. (4)

(c) What do you understand by terms machine and mechanism? State how these differ from each other. (3)

BE II YEAR III SEM. MID TERM-2

3ME45A(ME): PRODUCTION MACHINE TOOLS

TIME: 01 HOUR **MM: 20**

Q1. Explain the following in detail:

(a) TIG Welding (b) Electron Beam Welding (2x5 = 10)

Q2. What is punch and die? Explain different type of dies. (5)

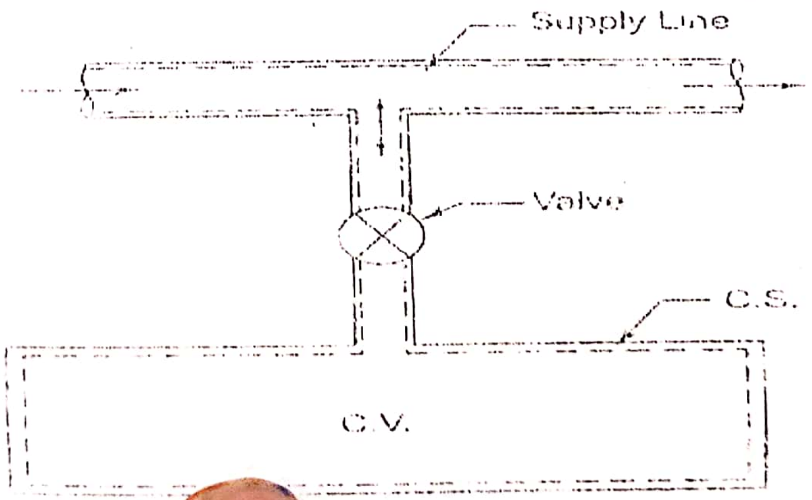
Q3. Differentiate between NC and CNC machine tools. Explain components of CNC machine. (5)

1st Midterm

ME 201A: Engineering Thermodynamics

Date 02/11/2023 Exam Duration: One Hour (12 PM to 1 PM) Maximum Marks: 20

1. State Kelvin-Plank and Clausius' statement. [2+2]
2. State entropy principle. [1]
3. Derive the expression of exergy balance for a closed system. [6]
4. What is stoichiometric air fuel ratio. What is excess air. Write combustion equation for fuel [illegible] [3]
5. A rigid, insulated tank/bottle is initially evacuated. A valve connected with the supply line is opened and the tank is charged with air (ideal gas) until the final pressure inside the tank reaches the pressure in the supply line. Derive the expression of final temperature inside the tank under [Final temperature = γ * temperature of the tank] [6]



2nd Midterm

ME 201A: Engineering Thermodynamics

Date 07/12/2023 **Exam Duration: One Hour (11 PM to 12 PM)** **Maximum Marks:20**

1. A single stage reciprocating compressor has a bore of 120mm and a stroke of 150mm, and is driven at a speed of 1200rpm. It is compressing CO_2 gas from a pressure of 120kPa and a temperature of 20°C to a temperature of 215°C. Assuming polytropic compression with n=1.3, no clearance and volumetric efficiency of 100%, calculate- (i) pressure ratio, (ii)Work done/cycle of compressor (iii)Indicated power (iii) shaft power with mechanical efficiency of 80%. **[6]**
2. Explain throttling process, joule kelvin coefficient, inversion curve. Plot T-P diagram showing inversion curve. Derive that for ideal gas Juole-kelvin coefficient is zero. **[7]**
3. Write the ideal gas equation and Vander-waal gas equation. Explain force of cohesion and co-volume terms. Explain Dalton's law of partial pressure, Amagat's law of additive volumes and Gibbs theorem. **[7]**

MBM University, Jodhpur

Department of Mechanical Engineering

B.E. IInd Year IIIrd Semester

First Mid-Term Examination Session 2023-24

ME-203A: Kinematics of Machines

Time: 1 hour **Maximum Marks – 20**

Q. 1(a). An open belt running over two pulleys of diameters 200 mm and 600 mm connects two parallel shafts placed at a distance of 2.5 m. The smaller pulleys rotates at 300 rpm and transmits 5 kW. The coefficient of friction between the belt and the pulley is 0.3. Determine: (i) length of belt; (ii) initial tension; (iii) minimum width if the safe working tension is 12 N/mm width. (6)

(b) Explain the following: (i) idler pulleys; (ii) intermediate pulleys; (iii) loose and fast pulleys; (iv) guide pulleys. (4)

Q. 2 (a) A multiple plate clutch transmits 55 kW of power at 1800 rpm. The coefficient of friction is 0.1. The inner radius is 80 mm and is 0.65 times the outer radius. If the intensity of pressure is not exceed 160 kN/m^2, determine the number of pair of friction surfaces needed to transmit the required torque. (6)

(b) What is friction circle? Derive an expression for its radius. (4)

MID TERM II (B.E. III SEM 2023-24)

ME:204a Mechanics of Solids

DEPT. OF MECH.ENGG., MBM UNIVERSITY

TIME: 60 MIN MM:30

Q1. A column 150 mm x 200 mm is 6 m long, both ends fixed E = $1.75 X 10^5$ N/mm^2.

a. Find Crippling load b. Safe load, if Fos = 3 (6)

Q2. At a point within a body subjected to two mutually perpendicular stresses 60 N/mm^2 and 30 N/mm^2 tensile and 45N/mm^2 of shear stress. Find σ_t, σ_n and σ_r on a plane inclined at 30° to normal cross-section. Solve with Mohr's circle method. (8)

Q3. Derive deflection equation (6)

OR

Q3. A hollow shaft is to transmit 300 KW at 80 rpm. If the sheer stress is not to exceed 60 Nlmm^2 and ID = 0.6 OD, find internal dia and external dia assuming minimum torque is 1.4 times the mean torque. Explain theory of pure bending and state the assumptions made. (6)

QUIZ : Write question no. and correct option in answer sheet. (10 =2x5)

		a.	b.	c.	d.
Q4	Rankine formula is good for	Short column	Long column	Medium column	Both 'a'and 'b'
Q5	Maximum slope in cantilever beam with UDL 'w' is	$wL^3/9EI$	$w L^3/16EI$	$w L^3/3EI$	$w L^3/6EI$
Q6	A column with minimum equivalent length has	both ends hinged	one end fixed and the other end free	one end fixed and other end hinged	both ends fixed
Q7	The load at which a vertical compression member just buckles is known as	Critical load	Crippling load	Buckling load	All of these
Q8	Which of the following statements is true for a simply supported beam?	Deflection at supports is maximum.	Deflection is maximum where slope is maximum.	Slope is minimum at supports	Deflection is maximum where slope is zero .

2nd Mid Term

3ME42(42): Material Technology

Date: 07/12/2023 **Exam Duration: 1 Hour (02:30 pm to 03:30 pm)** M.M.: 20

Question 1.

(A) Explain recovery, recrystallisation and grain growth in metals. (5)

(B) What is meant by hot and cold deformation in metals? (5)

Question 2.

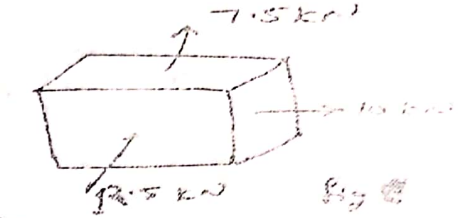
(A) What is the importance of heat treatment for metals. (4)

(B) Explain term annealing, tempering, normalizing and hardening. (6)

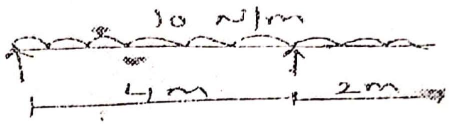
MID TERM I (B.E. III SEM 2023-24)
ME:204a Mechanics of Solids
MECH.ENGG. DEPT., MBM UNIVERSITY

[illegible] MIN MM:[illegible]

[illegible] in volume of the cube of 100X100 mm as shown in fig if 1/m=0.3 [illegible] $\times 10^5$ N/mm^2. (6)



[illegible] SFD and BMD for the beam shown:



[illegible] theory of pure bending and state the assumptions made. (6)

[illegible] question no. and correct option in answer sheet. (10 =2x5)

	a.	b.	c.	d.
If internal resistive force is 'F' and 'A' is resisting area, [illegible] is defined as	F/A	F	A/F	AxF
[illegible] out odd term	Young's modulus	Compressive stress	Shear stress	Tensile stress
[illegible] change in volume is ΔV and original volume is V, Volumetric Strain is defined as	ΔV/V	ΔV + V	V - ΔV	ΔV x V
[illegible] extension of a taper rod of length 'L' and tapering [illegible] diameters 'D_1' and 'D_2', subjected to a load (P), is	$4PL/\pi D_1D_2$	$PL/\pi D_1D_2$	$8PL/\pi D_1D_2$	$PL/4\pi D_1D_2$
[illegible] of a bar of uniform cross section 'A' of length [illegible] to self weight 'W' is given by	WL/2AE	WL/4AE	WL/8AE	WL/AE

BE II YEAR III SEM. MID TERM-1

3ME45A(ME): PRODUCTION MACHINE TOOLS

TIME: 01 HOUR **MM: 20**

Q1. Explain Constructional details and operations on

(a) Centre lathe (b) Universal milling machine (2x5 = 10)

Q2. Differentiate between Jigs and Fixture. Write the principles of location. (5)

Q3. Explain MIG welding with sketch. Differentiate between SMAW and MIG welding. (5)

MBM UNIVERSITY

MECHANICAL ENGINEERING DEPARTMENT

Mid Term Assessment – 02 November, 2023

Subject Code: 3ME42A(ME) **Subject: Materials Technology** **Max. Marks: 20**

1. What is difference between amorphous and crystalline materials? What are the various types of crystalline structures. Draw neat diagram of each. (7)
2. Explain and formulate following with respect to unit cell. (7)
   a) Number of atoms per unit cell.
   b) Atomic radius
   c) Atomic packing factor.
3. Explain various types of imperfection in crystals. Explain types of dislocations in crystal materials with suitable diagram. (6)

MBM University, Jodhpur

Department of Mechanical Engineering

B.E. IInd Year IIIrd Semester

Second Mid-Term Examination Session 2023-24

ME-203A: Kinematics of Machines

Time: 1 hour **Maximum Marks – 20**

Q. 1(a). In a four bar mechanism ABCD, the link AD is fixed link of 120 mm long, crank AB is 30 mm long and rotate uniformly at 100 rpm clockwise, while the link CD is 60 mm long and oscillates about fixed point D. The link BC is equal to length AD. Find the angular velocity of link DC when angle BAD is 60°. (6)

(b) What do you mean by instantaneous centre? Explain the properties of instantaneous centre. (4)

→ Rolling, slider, turning → belts, chain, rope

Q. 2 What is the difference between lower and higher pairs? Give examples of each type. (3)

(b) Sketch and explain the various inversion of single-slider crank chain mechanism. (4)

(c) What do you understand by terms machine and mechanism? State how these differ from each other. (3)

$\omega_{ab} = \frac{2\pi N}{60} = 4.17$ rad/s

m/s

lathe, planer

$v_b = \omega \times r$